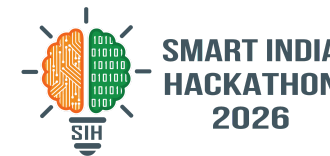


SMART INDIA HACKATHON 2026



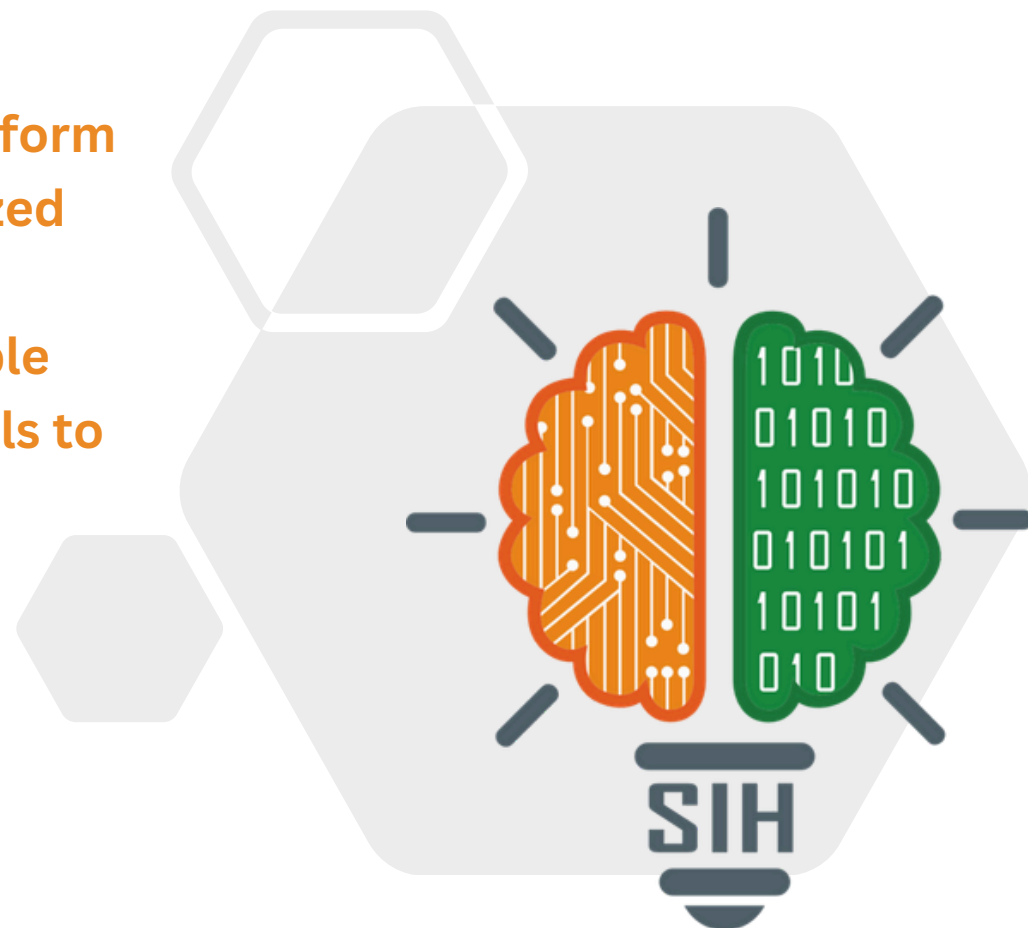
Problem Statement ID – **SIH26101**

Problem Statement - **Develop an AI enabled learning platform that identifies competency gaps, recommends personalized training through integration with the iGOT Karmayogi ecosystem, and capable of generating Quizzes and Multiple choice questions (MCQs) from uploaded learning materials to strengthen capacity building in India's Official Statistical System.**

Theme - **Smart Education**

Category - **Software**

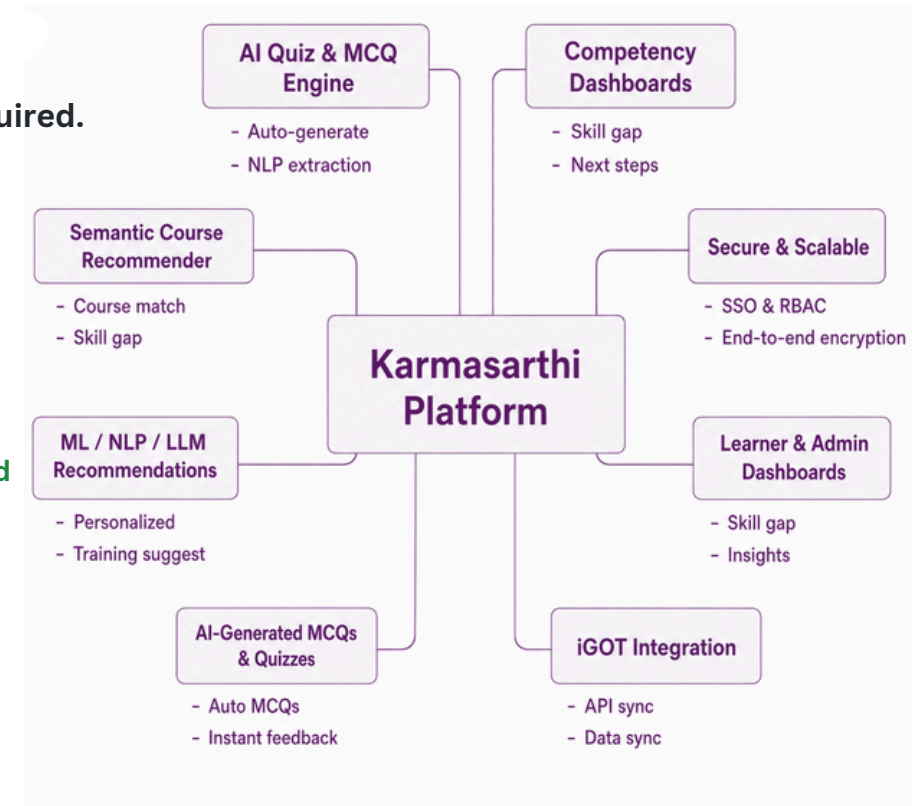
Team Name - **De-bugs_**





Meet the **FIRST SYSTEM** powered by **AGENTIC AI** that doesn't wait for a training-needs survey – it **autonomously** tracks every official's evolving profile, **maps** competency gaps in real time, **predicts** emerging skill demands before they become critical, and **orchestrates** personalized upskilling automatically. **A LIVING, ADAPTIVE WORKFORCE-INTELLIGENCE SYSTEM**. No manual gap analysis required.

- **AI Quiz & MCQ Engine**: Generates assessments from uploaded resources. Uses NLP to extract key concepts, generates **questions with adaptive difficulty, and provides contextual feedback**.
- **Semantic Course Recommender**: Analyzes skill gaps and learning history to surface highly-relevant recommendations. Integrates **NSSTA frameworks + MoSPI skill matrices** for evidence-based course ranking.
- **Competency Dashboards**: Individual dashboards show **competency radar, learning velocity, and recommended next steps**. Admin dashboards aggregate **org-wide gaps by department/role** with drill-down capabilities and **at-risk flagging**.
- **Secure & Scalable**: **Role-based access control, end-to-end encryption**, and cloud-ready deployment. Immutable audit trails, data residency compliance, differential privacy for analytics, DDoS protection.
- **Personalized Roadmap & Progress Tracking**: Generates **role-based learning roadmaps** from identified skill gaps and recommendations, then continuously **tracks course completion and competency progress**.



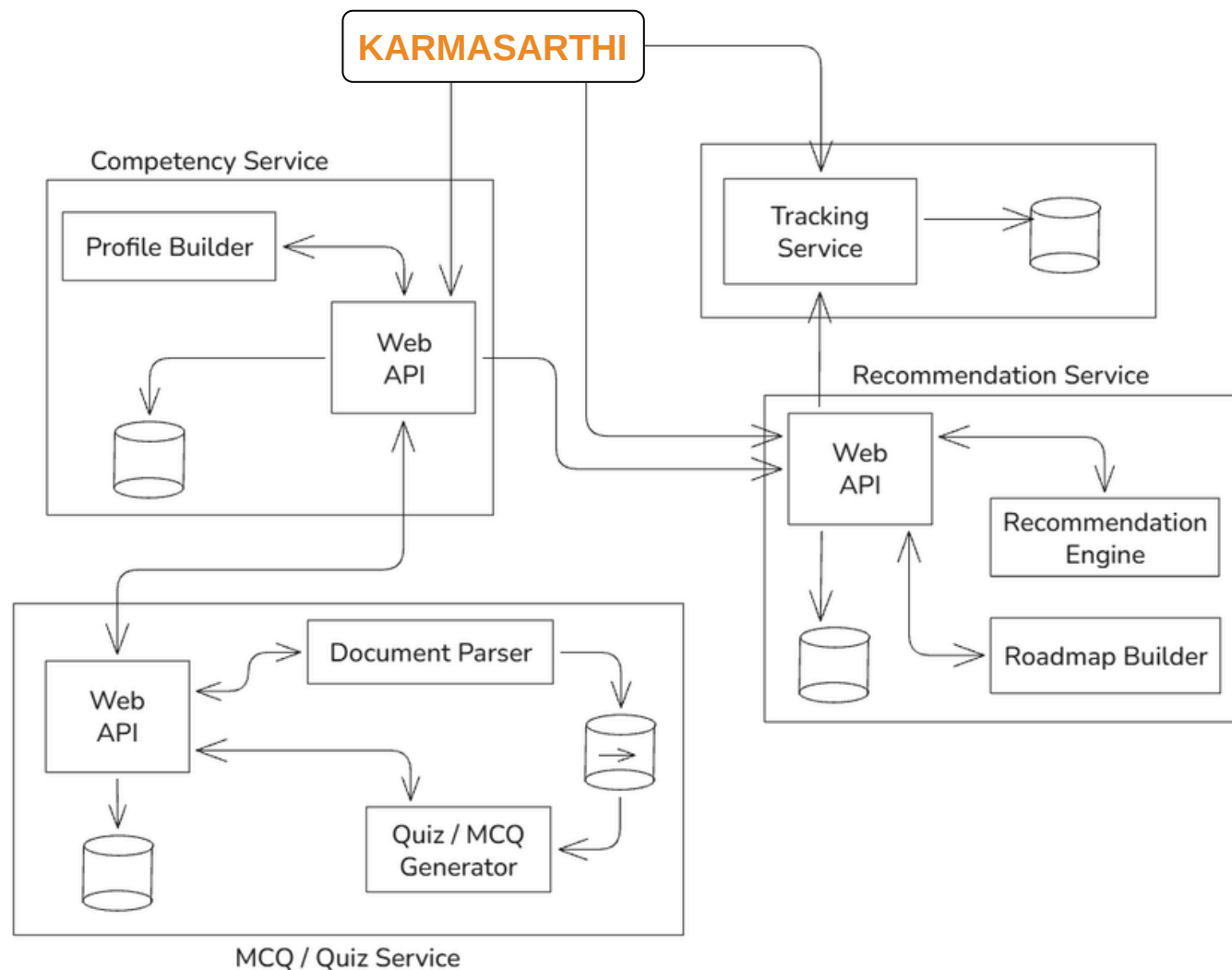
TECHNICAL APPROACH

Tech Stack



- REST adapter layer simulating **iGOT APIs** (course catalogue, enrolment, completion sync) - Scaled to actual data once provided
- Production integration via official iGOT API access upon **NIC / DoPT approval**
- Actual dataset provision from : **NSSTA & MoSPI**

System Architecture



FEASIBILITY AND VIABILITY

Feasibility

- **Proven:** Web, mobile & LLM recommendation/NLP engine with semantic search + real-time iGOT API sync
- **Phased build:** Profile → Gap Analysis → Personalised Pathways → AI Assistant → Workforce Analytics
- **Integration-ready:** iGOT Karmayogi + NSSTA infrastructure
- **AI-Ready Infrastructure** Adaptive assessments, auto-quizzes & multilingual NLP
- **Challenges:** API access, SSO, security & scalable semantic search

Viability

- **Govt-Funded Model:** Capacity-building infrastructure backed by MoSPI/DIID
- **Built-in user base:** Official Statistics cadre with adoption driven by ACPB mandates
- **Scalable:** Division Pilot → Pan-India MoSPI → Cross-Ministry
- **Sustained Engagement:** Continuous competency tracking linked to career progression

Use Cases

- **Officials:** Role-based, adaptive learning journeys + AI career assistant
- **DIID Admins:** Workforce competency & gap analytics
- **NSSTA Trainers:** Auto-generated multilingual assessments from training content
- **MoSPI Leadership:** Real-time workforce readiness dashboard

National Impact

- **Future-Ready Workforce:** Build digitally skilled statistical professionals
- **Predictive Skill Intelligence:** Spot emerging capability gaps before they become critical
- **Cross-Ministry DPI:** Reusable competency + recommendation engine across government
- **Smarter iGOT:** Drive relevance & outcomes, not just content consumption

Solution Architecture

- **Rollout:** Prototype → Pilot → MoSPI Scale → Cross-Ministry Framework
- **AI Layer:** Embeddings + LLMs + Adaptive Assessment + Virtual Assistant
- **Interoperability:** API-first integration with iGOT, NSSTA & future GovTech ecosystems
- **Security:** RBAC + SSO + Government Cybersecurity & Privacy Compliance

UPLOAD SURVEY / METHODOLOGY FILE:

Click to select a file or drag and drop here
Supported formats: PDF (.pdf), Word (.docx), Plain Text (.txt) up to 15MB

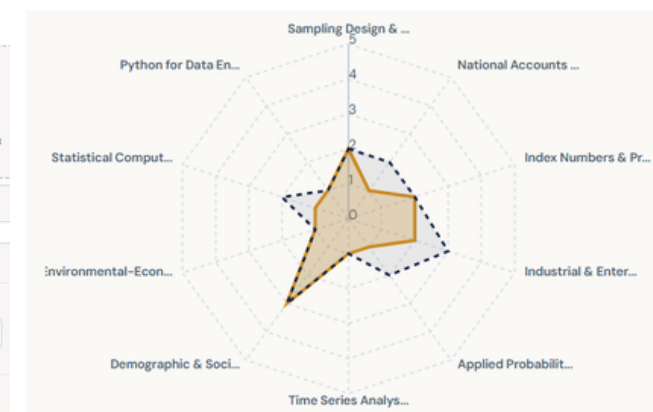
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GENERATION PARAMETERS & ALIGNMENT

FRAC Competency Domain: Statistical Competencies

Target Difficulty: Medium (Apply / Analyze)

☐ Force Deterministic Offline Fallback (Air-gapped field training mode without Gemini API)



IMPACT & BENEFITS

Autonomous / Proactive - Powered by **Agentic AI competency intelligence**, the system actively tracks profiles, surfaces gaps, and drives learning engagement – reducing the friction of manual training-needs assessment.

Benefits

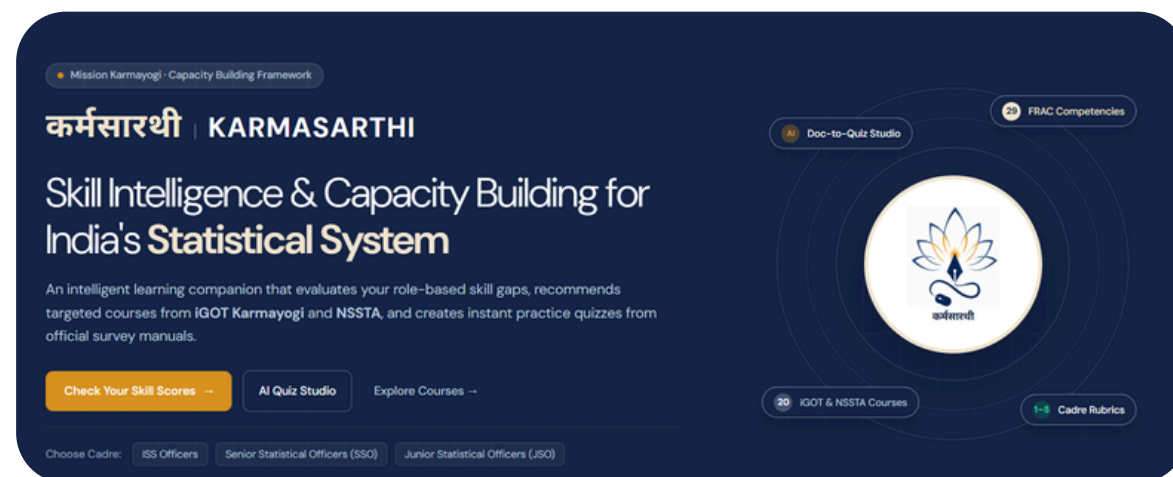
- **Individual:** Personalized, role-aligned upskilling instead of generic course browsing
- **Organizational (DIID/MoSPI):** Data-driven workforce planning via real-time competency visibility
- **Governance:** Directly supports Mission Karmayogi's shift toward measurable, appraisal-linked capacity building
- **Economic:** Better-targeted training reduces wasted training hours and repeated/irrelevant course enrollment
- **Technological:** Demonstrates practical AI/NLP adoption within core government digital infrastructure (DPI).

Potential Impact

- **Officials:** Faster, more relevant path from "assigned training" to "actually useful training"
- **DIID Admins:** Real-time visibility into workforce readiness instead of periodic manual surveys
- **NSSTA Trainers:** Reduced manual effort in assessment creation via AI-generated quizzes
- **MoSPI Leadership:** Earlier signal on emerging skill shortages (AI/ML, GIS, data governance) across the cadre

Supporting Facts

- **National Scale:** iGOT has 1.7+ crore users and 5,600+ courses across competencies.
- **Mandatory Adoption:** iGOT courses are now linked to APARs for Central Government & AIS officers, driving adoption.
- **AI-Ready Platform:** iGOT already uses AI Tutor, AI Sarthi & AI-CBP for personalised learning and competency mapping.
- **Proven Usage:** 7.3+ crore enrolments and 3.35+ crore completions provide real-world data for recommendation models.
- **Strong Demand:** 94% of organisational leaders report AI competency gaps, validating the need for AI-driven skill-gap analysis.



REFERENCES

Github Repository

<https://github.com/Rewant-1/sih-26>

RESEARCH AND REFERENCES

Comparable Platforms:

- Degreed — Skills & learning platform
- Docebo — AI-powered LMS
- iGOT Karmayogi itself

Research & Best Practices:

- WEF Future of Jobs Report 2025 — Global AI skills gap data
- SHRM 2025 Talent Trends — AI adoption in L&D
- PIB / DD News — iGOT usage & policy updates
- Mission Karmayogi Framework — Competency-based capacity building

Comparison with existing systems

Feature	Our Platform	Karmayogi	Degreed	Docebo
Course repository	✓ (via iGOT + NSSTA)	✓ (5,600+ courses)	✓ (aggregated)	✓
AI-based skill-gap analysis	✓	Partial	✓	✓
Personalized recommendations	✓	✗	✓	✓
Official-Statistics-specific competency	✓	✗	✗	✗
NSSTA TPAC integration	✓	✗	✗	✗
AI MCQs/quizzes from uploaded material	✓	✗	✗	Partial
Admin analytics dashboard	✓	✓ (general)	✓	✓
Government data-privacy/SSO alignment	✓ (designed)	✓ (native)	✗	✗